Special Instructions: You must write your name and netid on every sheet of the exam.

ECE 340 – Spring 2026

Name_____

Net ID#_____

Section_____

ECE 340 – EXAM 1
Thursday March 5th, 2026 7:00 – 8:00 p.m.

1002 ECEB A (He), C (Bogdanov), Sections X (Bayram)

NOTE: This is a closed book and closed notes exam. The exam consists of three problems. Unless stated otherwise, do your work on the page of the problem and if necessary on the preceding blank page. A simple scientific calculator is allowed but it must have no wireless or internet connectivity. Cell phones are not permitted.
Be sure to explicitly show the units in your work, as well as in your answers. Circle your answer. Be neat!

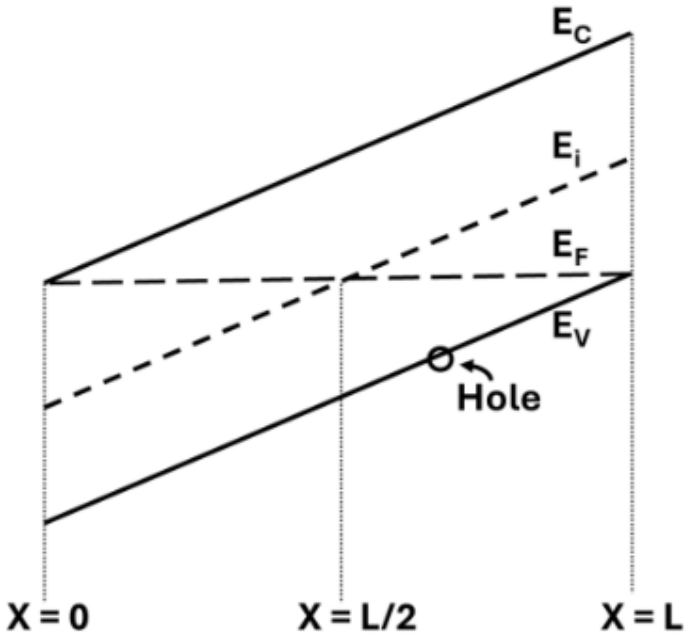
If we cannot read or follow your work, you get zero credit!

For each problem, you must show complete work and **indicate your reasoning**. No credit will be given if you do not show the complete work *and describe your procedure*, even if the answer is correct. Write your name, ID#, and section on this page and **sign below**.

	Prob.#1		Prob.#2		Prob.#3	
	33 pts.		33 pts.		34 pts.	
a	8		10		8	
b	10		13		7	
c	6		10		7	
d	9		---		12	
e	---		---		---	
f	---		---		---	
g	---		---		---	
T	33		33		34	

Signature:_____ Your Exam Score:_____

Problem 1



A semiconductor under equilibrium conditions is characterized by the energy band diagram given above. Bandgap of the semiconductor (E_G) is 1.2 eV, $n_i = 10^{11}$ /cm^3, kT = 0.026 eV.

a) At X = L/2, what are the i) electron concentration n(X) for X = L/2 and ii) hole concentration p(X) for X = L/2? [8 points]

i) $n = n_i = 10^{11}$ cm^{-3}

ii) $p = n_i = 10^{11}$ cm^{-3}

b) At X = L/4, i) is the material n-type or p-type? ii) what is the electron concentration n(X=L/4)? [10 points]

i) E_F is closer to E_c. n-type.

ii) $n = n_i \times \exp((E_F - E_i)/kT) = 10^{11} \times \exp(0.3/0.026) \approx 1 \times 10^{16}$ /cm^3

Name: _____ Net ID: _____

c) What is the approximate kinetic energy of the hole shown in the diagram? [6 points]

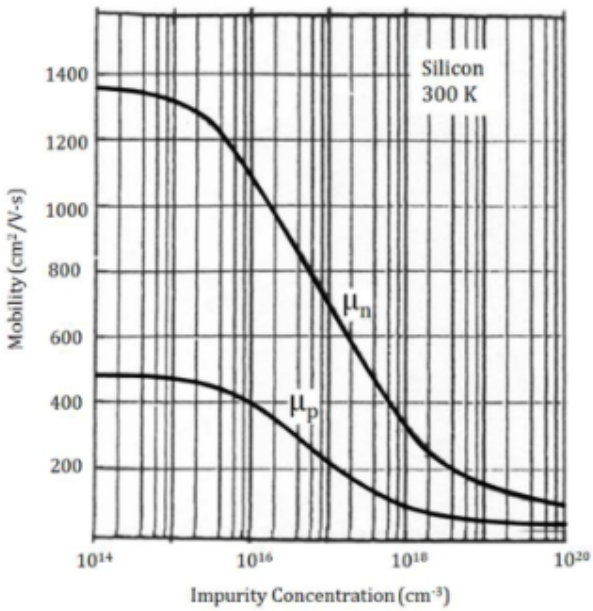
KE ~ 0 since the hole is approximately sitting at the valence band edge.

d) If L = 10 μm, what is the magnitude of the electric field inside the semiconductor? [9 points]

E-field $= \frac{1}{q} \times \frac{dEi}{dx} = \frac{1}{q} \times \frac{E_g}{L} = \frac{1}{q} \times \frac{1.2eV}{10 \times 10^{-4} cm} =$ 1200 V/cm

Problem 2

Consider two silicon samples at temperature 300K. Sample one (**S1**) is doped with acceptors with a concentration $N_a = 1 \times 10^{17}$ cm^{-3}. Sample two (**S2**) is doped both with acceptors having a concentration $N_a = 1 \times 10^{17}$ cm^{-3} and with donors at a concentration $N_d = 5 \times 10^{17}$ cm^{-3}. The electron and hole mobility *vs.* impurity concentration is shown in the figure below. The intrinsic carrier concentration for silicon n_i at 300K is 1.5×10^{10} cm^{-3}.



a) Calculate the conductivity σ (Ω-cm)$^{-1}$ of sample **S1**. [10 points]

Solution: $p \approx N_a = 10^{17} cm^{-3} \gg n_i$

At impurity level of 10^{17} cm^{-3}, $\mu_p = 200\ cm^2/Vs$

$$\sigma \approx qp\mu_p = 1.6 \times 10^{-19} \times 10^{17} \times 200 = 3.2\ (\Omega \cdot \text{cm})^{-1}$$

b) Calculate the conductivity σ (Ω-cm)$^{-1}$ of sample **S2**. Specify the doping concentration, the value of majority carrier mobility, and the impurity concentration you used to determine the mobility value. [13 points]

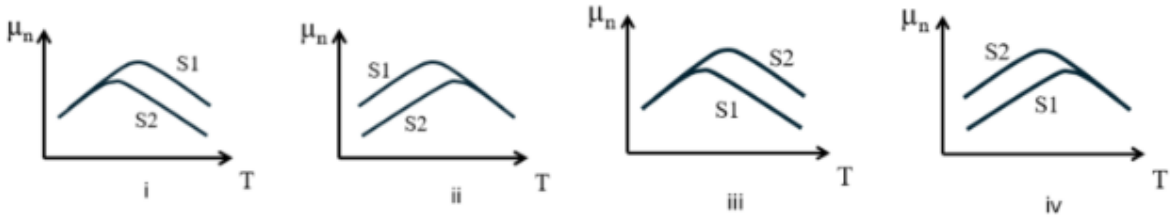
Solution: Doping concentration: Since $N_d > N_a$, thus sample is n doped and majority carrier is electrons, with $n \approx N_d - N_a = 5 \times 10^{17} cm^{-3} - 1 \times 10^{17} cm^{-3} = 4 \times 10^{17} cm^{-3}$

Impurity concentration = $N_a + N_d = 1 \times 10^{17} cm^{-3} + 5 \times 10^{17} cm^{-3} = 6 \times 10^{17} cm^{-3}$

Use impurity concentration of $6 \times 10^{17} cm^{-3}$, one found $\mu_n = 400 cm^2/Vs$

$$\sigma = qn\mu_n = 1.6 \times 10^{-19} \times 4 \times 10^{17} \times 400 = 25.6\ (\Omega \cdot cm)^{-1}$$

c) Consider figure *i)* through figure *vi)* as shown below. Which figure correctly illustrates the **electron mobility** *vs.* temperature relation for sample S1 and sample S2? Briefly justify your answer. (10 points)



Solution: Figure ii is correct. In lower temperature regime, the dominant scattering mechanism is impurity scattering, which is predominantly affected by impurity levels. The higher the impurity level, the lower the mobility. Sample S2 has higher impurity level than sample S1. Thus, sample S2 should have lower mobility than sample S1 at low temperature regime. For higher temperature regime, the dominant scattering mechanism is lattice scattering, which should be identical for both sample S1 and sample S2. Thus, figure ii is correct.

Problem 3

A uniformly doped n-type silicon sample at T = 300K has the following properties:

Doping concentration	$N_D = 1.5 \times 10^{16}\ cm^{-3}$
Carrier lifetimes	$\tau_n = \tau_p = 15\ \mu s$
Electron mobility	$\mu_n = 1350\ cm^2 \cdot V^{-1} \cdot s^{-1}$
Hole mobility	$\mu_p = 450\ cm^2 \cdot V^{-1} \cdot s^{-1}$
Intrinsic concentration	$n_i = 1.5 \cdot 10^{10}\ cm^{-3}$

A 10% numerical accuracy is acceptable in this problem.

a) The sample is illuminated uniformly with a constant optical generation rate $G = 10^{19}\ cm^{-3}s^{-1}$. Calculate the steady-state excess electron and hole concentrations δn and δp. Is this a low-level or high-level injection? Justify your answer briefly. [8 points]

At equilibrium, $n_0 \approx N_D = 10^{16}cm^{-3}$. Under uniform steady-state illumination, the excess concentration is given by $\delta n = \delta p = G\tau_p = 10^{19}cm^{-3}s^{-1} \cdot 1.5 \cdot 10^{-5}s = \boxed{1.5 \cdot 10^{14}cm^{-3}}$. Because excess concentration is far lower than the majority carrier concentration, this situation is considered a low-level injection.

b) The illumination is suddenly turned off at $t = 0$. Write the expression for the total hole concentration for all $t > 0$ as a function of time. Plugging in the numerical values is optional. [7 points]

After the light is turned off, excess carriers recombine with lifetime τ_p, until the equilibrium concentration is attained

$$\boxed{p(t) = p_0 + \delta p(t=0)e^{-\frac{t}{\tau_p}}}$$

The equilibrium hole concentration is $p_0 = \frac{n_i^2}{N_D} = \frac{(1.5 \cdot 10^{10}cm^{-3})^2}{1.5 \cdot 10^{16}cm^{-3}} = 1.5 \cdot 10^4 cm^{-3}$

$$\boxed{p(t) = 1.5 \cdot 10^4 cm^{-3} + 1.5 \cdot 10^{14}cm^{-3}e^{-\frac{t}{1.5 \cdot 10^{-5}s}}}$$

Name: _____ Net ID: _____

c) For the steady-state condition of question a), calculate the quasi-Fermi levels for holes relative to the intrinsic Fermi level (F_p - E_i). Express your answers in meV. [7 points]

The hole Quasi-Fermi level is defined by and $p = n_i \, exp\left(\frac{E_i - F_p}{kT}\right)$.
For the holes, the quasi-Fermi level is located at:

$$F_p - E_i = -kT \, ln\left(\frac{p}{n_i}\right) = 25\text{meV} \, ln\left(\frac{1.5 \cdot 10^{14}\text{cm}^{-3}}{1.5 \cdot 10^{10}\text{cm}^{-3}}\right) = \boxed{-230 \text{ meV}}$$

d) The illumination is changed such that at $t = 0$ a gradient in the excess carrier concentration is produced that follows a linear profile from $x = 0$ to $x = 1\,\mu m$. Assuming $\delta n(x = 0) = \delta p(x = 0) = 10^{14} cm^{-3}$ and $\delta n(x = 1\,\mu m) = \delta p(x = 1\,\mu m) = 10^{16} cm^{-3}$ and that there is no electric field, calculate the electron and hole diffusion current densities at $x = 0.5\,\mu m$. [12 points]

First, we obtain the diffusion coefficients from the Einstein relation:

$$D_p = \mu_p \frac{kT}{q} = 450\text{cm}^2 \cdot V^{-1} \cdot s^{-1} \cdot 0.025V \approx 11\text{cm}^2 \cdot s^{-1}$$

$$D_n = \mu_n \frac{kT}{q} = 1350\text{cm}^2 \cdot V^{-1} \cdot s^{-1} \cdot 0.025V \approx 34\text{cm}^2 \cdot s^{-1}$$

The carrier concentration gradient at $x = 0.5$ µm are:

$$\left.\frac{dp}{dx}\right|_{x=0.5\,\mu\text{m}} = \left.\frac{dn}{dx}\right|_{x=0.5\,\mu\text{m}} \frac{10^{16}\text{cm}^{-3} - 10^{14}\text{cm}^{-3}}{10^{-4}\text{ cm}} \approx 10^{20}\text{cm}^{-4}$$

Thus, we calculate the hole diffusion current as:

$$J_{\text{p,diff}}(x = 0.5\,\mu\text{m}) = -qD_p \left.\frac{dp}{dx}\right|_{x=0.5\,\mu\text{m}} = 1.6 \cdot 10^{-19}C \cdot 11\text{cm}^2 \cdot s^{-1} \cdot 10^{20}\text{cm}^{-4}$$

$$\approx \boxed{-180\text{ A} \cdot \text{cm}^{-2}}$$

Analogously, the electron diffusion current is:

$$J_{\text{n,diff}}(x = 0.5\mu\text{m}) = qD_n \left.\frac{dn}{dx}\right|_{x=0.5\,\mu\text{m}} = 1.6 \cdot 10^{-19}C \cdot 34\text{cm}^2 \cdot s^{-1} \cdot 10^{20}\text{cm}^{-4}$$

$$\approx \boxed{+540\text{ A} \cdot \text{cm}^{-2}}$$